**1.In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.**

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

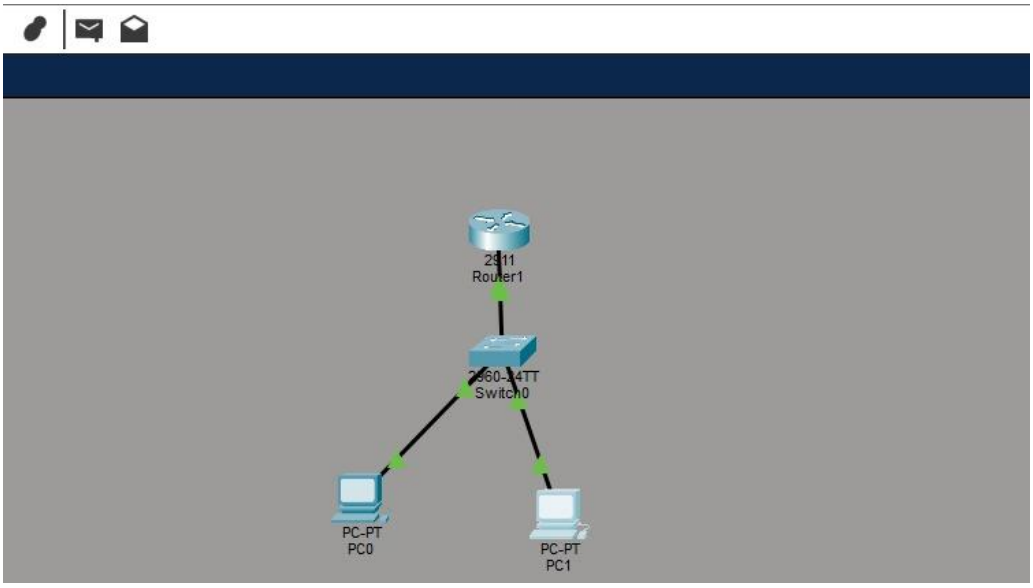
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

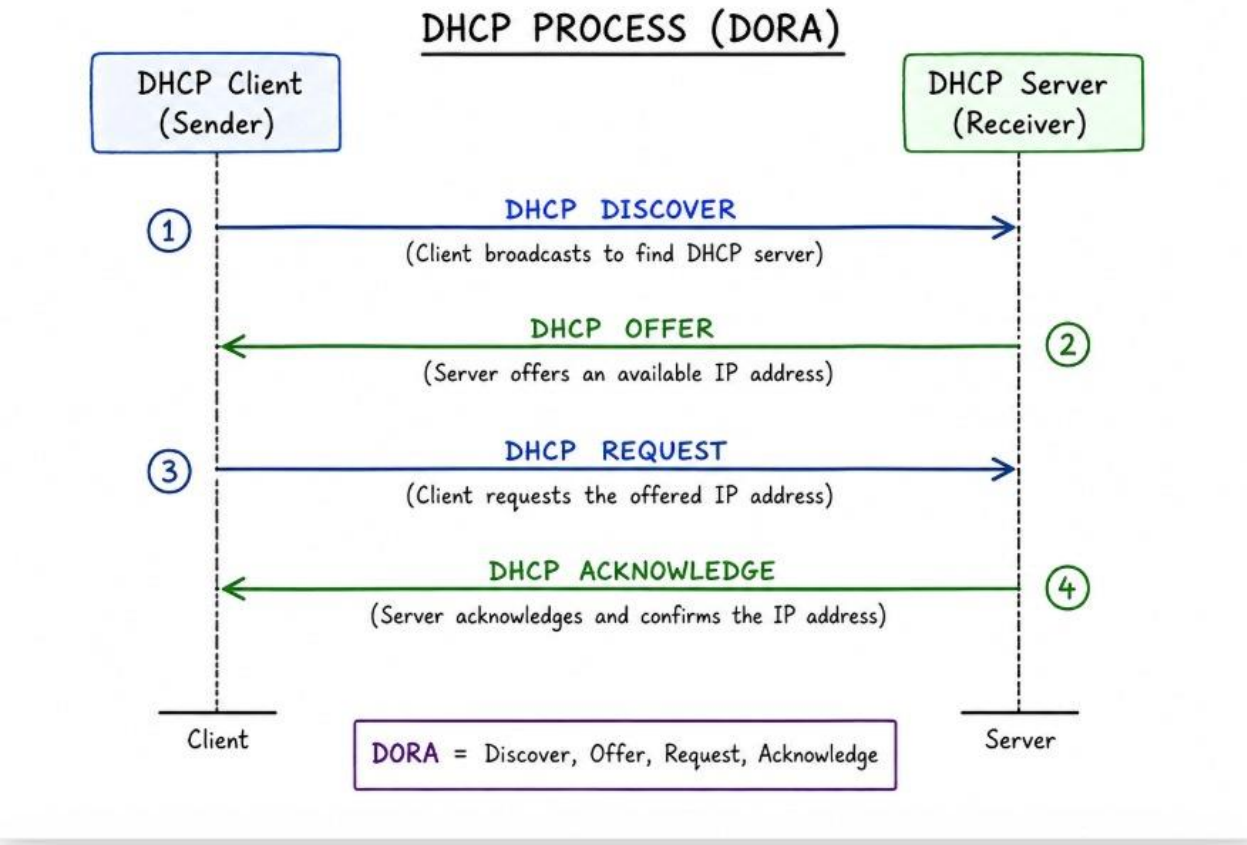
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.9
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-roter 192.168.1.1
                   ^
% Invalid input detected at '^' marker.

Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```



**2.Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.Hint: Use arrows to show the direction of each message between client and server.**



**3.Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.**

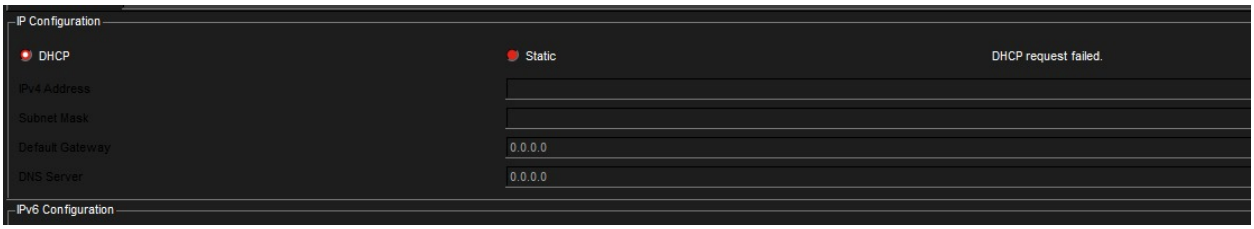
```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut down

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.9
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network  192.168.1.1 255.255.255.0
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.254
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network  192.168.1.1 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```



**4.Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.**

Devices: Router (DHCP enabled), Laptop, and Smartphone .

- Network connection problem – The device is not properly connected.
- DHCP server is OFF – The server may be down.
- DHCP service is OFF – DHCP service may not be enabled.
- DHCP scope is full – All available IP addresses are already used.
- Wrong DHCP configuration – IP range, subnet mask, or gateway may be incorrect.
- Router or switch problem – The device may not be able to reach the DHCP server.